

METIMMOX-1 Expected Values of Partially Perfect Information (EVPPI) Analysis

Biomarker-Guided Strategies for Selecting Immunotherapy Candidates in Metastatic MSS/pMMR Colorectal Cancer

Ben Geisler

2025-10-15

Table of contents

1	Introduction	1
2	Methods	1
2.1	EVPPI Calculation Methodology	1
2.2	Implementation	2
3	Results	3
3.1	Decision Uncertainty Analysis	3
3.2	EVPPI Results	4
4	Discussion	4
4.1	Interpretation of Results	4
4.2	Limitations	4
5	Conclusions	4

1 Introduction

This report presents the Expected Values of Partially Perfect Information (EVPPI) analysis for the METIMMOX-1 economic model. EVPPI quantifies the maximum value of obtaining perfect information about specific model parameters or groups of parameters, helping to prioritize future research efforts.

The EVPPI analysis identifies which parameters, if known with certainty, would provide the greatest reduction in decision uncertainty. This information is crucial for:

- Prioritizing future research investments
- Identifying parameters requiring more precise estimation
- Guiding the design of additional studies
- Supporting evidence development strategies

2 Methods

2.1 EVPPI Calculation Methodology

The EVPPI for parameter θ is calculated as:

$$EVPPI_{\theta} = E_{\theta}[max_j E_{\theta^{-\theta}|\theta}[NMB(j, \theta, \theta^{-\theta})]] - max_j E[NMB(j, \theta)]$$

Where: - θ represents the parameter(s) of interest - θ^{-} represents all other parameters
- j represents treatment strategies - NMB represents Net Monetary Benefit

2.2 Implementation

The EVPPI analysis uses a nested sampling approach with diagnostic capabilities to understand decision uncertainty patterns.

3 Results

3.1 Decision Uncertainty Analysis

****Decision Uncertainty Assessment:****

- Total EVPI: € 0
- Expected value with perfect information: € 70292.51
- Expected value with current information: € 70292.51
- Optimal strategy: control
- Proportion of simulations favoring optimal strategy: 100 %
- Decision uncertainty index: 0

****Interpretation:**** The EVPI of € 0 indicates very low decision uncertainty. The optimal st

Table 1: Strategy Performance Summary

	Strategy	Times Optimal	Percentage (%)	Expected NMB (€)
	control	control	5000	100
				70292.51

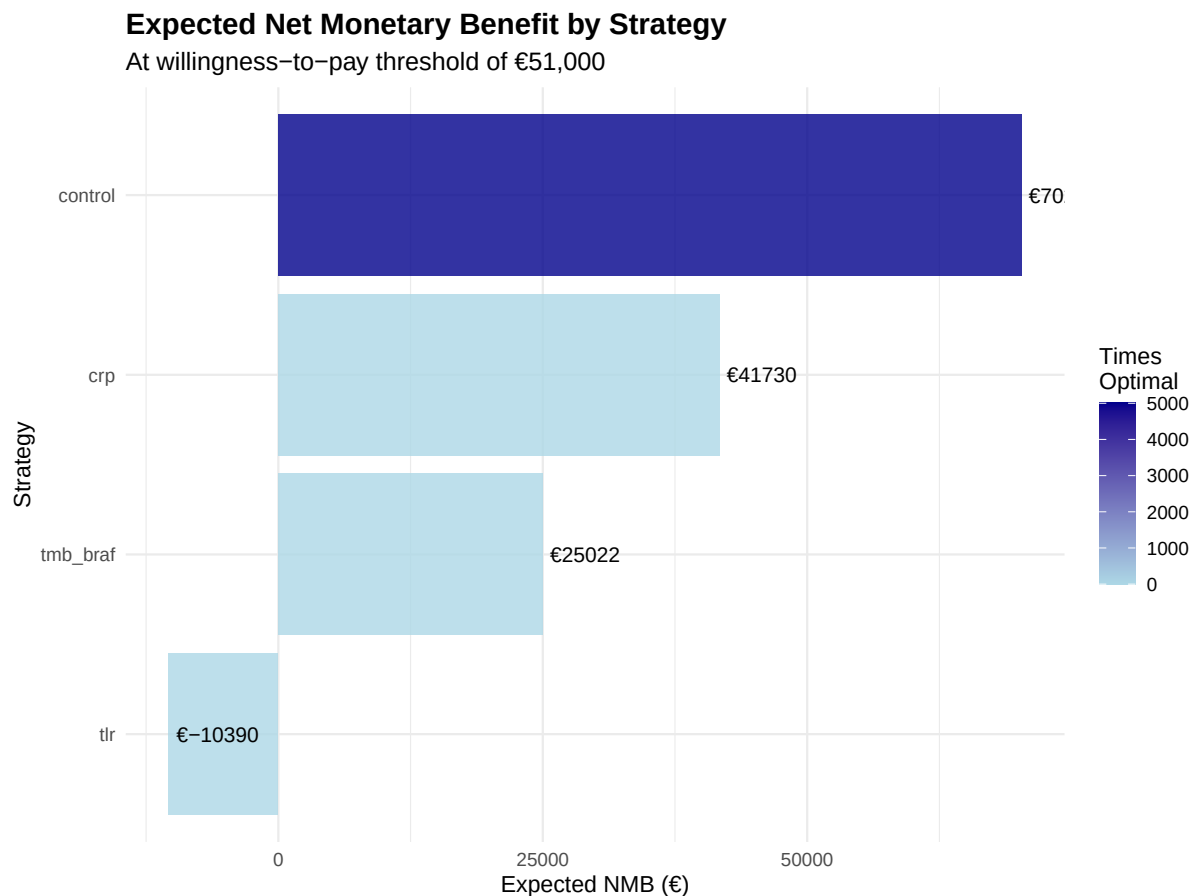


Figure 1: Net Monetary Benefit by Strategy

Table 2: Expected Values of Partially Perfect Information by Parameter

Parameter	EVPI (€)	% of EVPI	Notes
Nivolumab cost	0	0	EVPI is zero or negative
FLOX cost	0	0	EVPI is zero or negative
NGS test cost	0	0	EVPI is zero or negative
CT scan cost	0	0	EVPI is zero or negative
Non-progression utility	0	0	EVPI is zero or negative
Progression utility	0	0	EVPI is zero or negative
End-of-life costs	0	0	EVPI is zero or negative
Visit costs	0	0	EVPI is zero or negative
Baseline costs	0	0	EVPI is zero or negative
Follow-up costs	0	0	EVPI is zero or negative

3.2 EVPPI Results

4 Discussion

4.1 Interpretation of Results

The analysis reveals the following key insights:

1. **Minimal Decision Uncertainty**: The EVPI of € 0 indicates minimal decision uncertainty.
2. **Dominant Strategy**: The ' control ' strategy is optimal in 100 % of simulations, demonstrating robustness.
3. **Limited Research Value**: The extremely low EVPI suggests that additional research to reduce uncertainty has minimal expected value.
4. **Robust Decision Framework**: The low decision uncertainty indicates that the biomarker-based decision framework is robust to parameter variations.

Clinical and Policy Implications:

- The optimal strategy appears well-established under current evidence.
- Decision uncertainty is not a barrier to implementation.
- Research priorities should focus on implementation rather than additional parameter estimation.

4.2 Limitations

The EVPPI analysis has several important limitations:

1. **Model Structure Dependence**: Results are conditional on the current model structure and assumptions.
2. **Parameter Range Assumptions**: EVPI calculations depend on the assumed uncertainty distributions for parameters.
3. **Linear Utility Assumption**: The analysis assumes linear utility in monetary outcomes.
4. **Implementation Considerations**: Low EVPI does not address implementation challenges or real-world effectiveness.

5 Conclusions

Key Findings:

- Total EVPI: € 0
- Optimal strategy: control
- Decision certainty: 100 %

****Recommendations:****

- ****Implementation Focus****: Given the low decision uncertainty, efforts should focus on implementation
- ****Real-World Evidence****: Future research should prioritize implementation studies and real-world evidence
- ****Alternative Interventions****: Research resources might be better allocated to evaluating alternative interventions
- ****Health System Considerations****: Focus on addressing barriers to implementation of the optimal strategy

Report compiled on 2025-10-15 by Ben Geisler using the METIMMOX-1 health economic model.